



SYMBIOSIS INSTITUTE OF DIGITAL AND TELECOM MANAGEMENT

CONSTITUENT OF SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

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Guesstimate Problem Statement

Estimate the Number of Spotify Premium Subscribers in India.

As a consultant, you are required to estimate the number of Spotify Premium users in India without using external data sources. Use logical assumptions, population segmentation, internet penetration, music consumption behaviour, and premium conversion rates to arrive at an approximate estimate.

Step-by-Step Solution

Step 1: Start with India's Population

Assume:

Population of India = 1.4 Billion

Step 2: Segment Population by Urban and Rural Areas

Assumption

Spotify Premium adoption is highly dependent on internet access and digital payment adoption. Therefore, we first divide the population into Urban and Rural segments.

Segment	% Population	Population
Urban	35%	490 Million
Rural	65%	910 Million
Total	100%	1.4 Billion

Step 3: Divide Population by Income Groups

People with higher disposable income are more likely to purchase premium subscriptions.

Urban Population (490 Million)

Income Group	Share
High Income	20%
Middle Income	40%
Low Income	40%

Rural Population (910 Million)

Income Group	Share
High Income	10%
Middle Income	40%
Low Income	50%

Step 4: Estimate Internet Penetration

Spotify can only be used by people with internet access.

Assumptions

Segment	Internet Penetration
Urban High Income	100%
Urban Middle Income	70%
Urban Low Income	30%
Rural High Income	50%
Rural Middle Income	20%
Rural Low Income	5%

Calculate Internet Users

Urban High Income

$$490 \times 20\% \times 100\%$$

= 98 Million

Urban Middle Income

$$490 \times 40\% \times 70\%$$

= 137.2 Million

Urban Low Income

$$490 \times 40\% \times 30\%$$

= 58.8 Million

Rural High Income

$$910 \times 10\% \times 50\%$$

= 45.5 Million

Rural Middle Income

$$910 \times 40\% \times 20\%$$

= 72.8 Million

Rural Low Income

$$910 \times 50\% \times 5\%$$

= 22.75 Million

Total Internet Users

$$98 + 137.2 + 58.8 + 45.5 + 72.8 + 22.75$$

= 435.05 Million

≈ 435 Million people

Step 5: Segment Internet Users by Age

Spotify usage is generally higher among younger demographics.

Assumption

Age Group	Share
0–14 Years	26%
15–24 Years	18%
25–34 Years	16%
35–44 Years	14%
45+ Years	26%

Step 6: Estimate Music Streaming Adoption

Not every internet user listens to music online.

Assume the following percentage of internet users in each age group are music listeners:

Age Group	Music Listener %
0–14	50%
15–24	90%
25–34	85%
35–44	60%
45+	40%

Music Listeners by Age Group

0–14 Years

$$435 \times 26\% \times 50\%$$

$$= 56.55 \text{ Million}$$

15–24 Years

$$435 \times 18\% \times 90\%$$

$$= 70.47 \text{ Million}$$

25–34 Years

$$435 \times 16\% \times 85\%$$

$$= 59.16 \text{ Million}$$

35–44 Years

$$435 \times 14\% \times 60\%$$

$$= 36.54 \text{ Million}$$

45+ Years

$$435 \times 26\% \times 40\%$$

$$= 45.24 \text{ Million}$$

Total Music Listeners

$$56.55 + 70.47 + 59.16 + 36.54 + 45.24$$

= 267.96 Million

≈ 268 Million Music Listeners

Step 7: Estimate Spotify Users

Assumption

Spotify is one of several music-streaming platforms in India competing with:

- YouTube Music
- JioSaavn
- Gaana
- Apple Music
- Amazon Music

Assume:

30% of all online music listeners use Spotify

Spotify Users

= $268 \times 30\%$

= **80.4 Million Users**

Step 8: Estimate Premium Conversion Rate

Not all Spotify users pay for Premium.

Premium conversion varies by age and purchasing power.

Age Group	Premium Conversion
0–14 Years	5%
15–24 Years	40%
25–34 Years	40%
35–44 Years	20%
45+ Years	10%

Premium Subscribers Calculation

0–14 Years

$$56.55 \times 30\% \times 5\%$$

$$= 0.85 \text{ Million}$$

15–24 Years

$$70.47 \times 30\% \times 40\%$$

$$= 8.46 \text{ Million}$$

25–34 Years

$$59.16 \times 30\% \times 40\%$$

$$= 7.10 \text{ Million}$$

35–44 Years

$$36.54 \times 30\% \times 20\%$$

$$= 2.19 \text{ Million}$$

45+ Years

$$45.24 \times 30\% \times 10\%$$

$$= 1.36 \text{ Million}$$

Step 9: Arrive at Final Estimate

Total Premium Subscribers

$$= 0.85 + 8.46 + 7.10 + 2.19 + 1.36$$

$$= 19.96 \text{ Million}$$

≈ 20 Million Spotify Premium Subscribers

Final Answer

Estimated Spotify Premium Subscribers in India ≈ 20 Million

Consulting Interview Takeaway

The interviewer is not evaluating whether the answer is exactly correct. They are evaluating:

1. **Structured Thinking** – Did you break the problem logically?
2. **MECE Segmentation** – Did you use mutually exclusive and collectively exhaustive buckets?
3. **Reasonable Assumptions** – Are your assumptions defensible?
4. **Mental Math Accuracy** – Can you calculate quickly and correctly?

5. **Business Logic** – Did you connect internet access, income, age, and willingness to pay?

A concise consulting framework for this guesstimate would be:

Population → Internet Users → Music Streamers → Spotify Users → Premium Conversion → Premium Subscribers